INTEGRATED SCHOOL MANAGEMENT SYSTEM

OUR LADY OF SAINT ALVAREZ

IDENTIFY THE KEY FUNCTION OF EACH SYSTEM:

Registar - Student Enrollment (new and current), Course Registration, Grade Management (recording and retrieval), Transcript Generation, Academic Status tracking.

Finance - Tuition & Fee Management (billing and payment tracking), Payroll Processing (for HR staff), Budgeting, Financial Reporting.

Library - Book & Resource Cataloging, Lending/Borrowing Tracking, Late Fee/Fine Management, User Authentication (checking student/staff status).

Human Resource - Staff/Faculty Records management, Time & Attendance, Benefits Administration, Recruitment and Onboarding.

Student Portal - Personal Information updating, Course Schedule viewing, Grade viewing, Fee Balance viewing, Library Account access, Communication (announcements).

Explain **WHY** and **HOW** these systems should be integrated:

- **WHY**
  To Eliminate Data Redundancy and Discrepancy
  To Automate Business Processes (Enrollment to Billing, Graduation Clearance)
  To Improve Student and Staff Experience
- **HOW**
  Registrar > Finance: Student enrollment and subjects for tuition billing
  Registrar > Library/HR: Student and staff status for account access
  Finance > Student Portal: Balance, payments, billing info Library > Finance: Book fines and penalties
  HR > Finance: Payroll details

Suitable type of **Integration:**

System integration - We think this is the most suitable because the school is connecting multiple separate systems (Registrar, Finance, Library, HR, Student Portal) to form one larger system/environment.

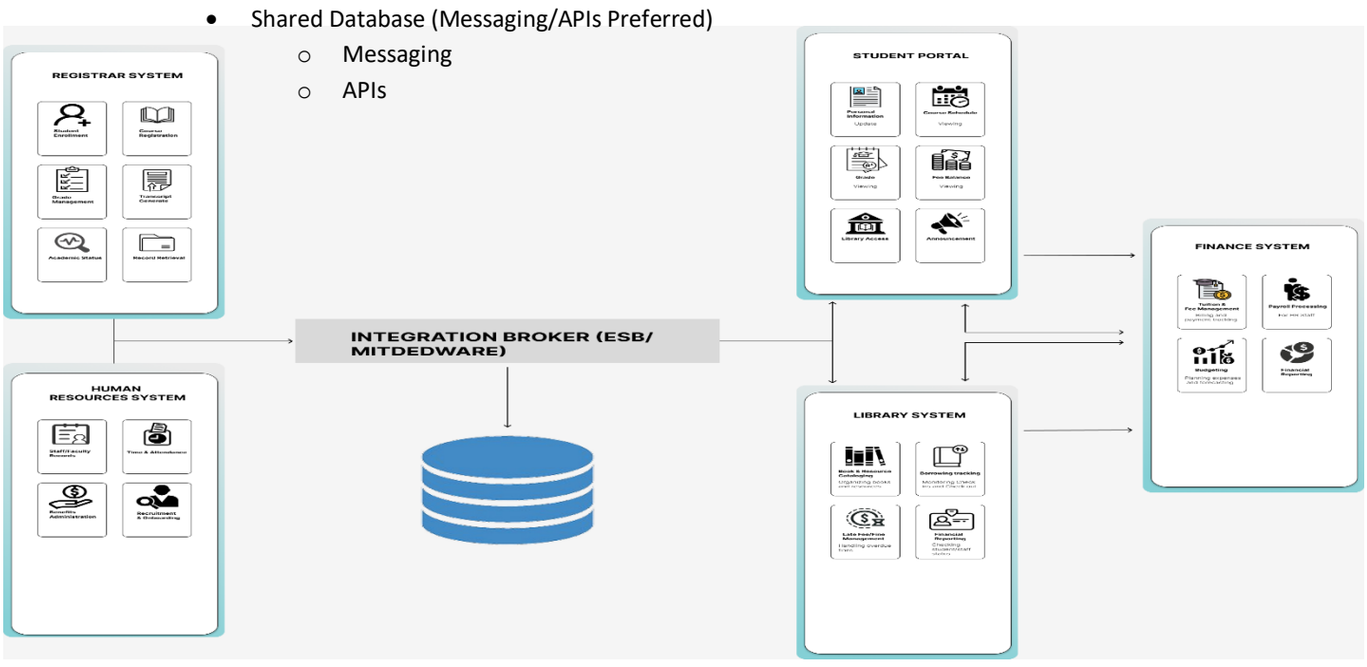
Benefits, Challenges, and **Technologies/Methods**

Benefits

- Improved Operational Efficiency
- Accurate and Real-Time Reporting
- Enhanced Data Integrity
- Better Service Delivery

Challenges

- - Data Heterogeneity (The Problem Middleware Solves)
- System Ownership and Maintenance
- Security
- Cost and Time

Technologies/Methods

- Middleware
- Shared Database (Messaging/APIs Preferred)
  - Messaging
  - APIs